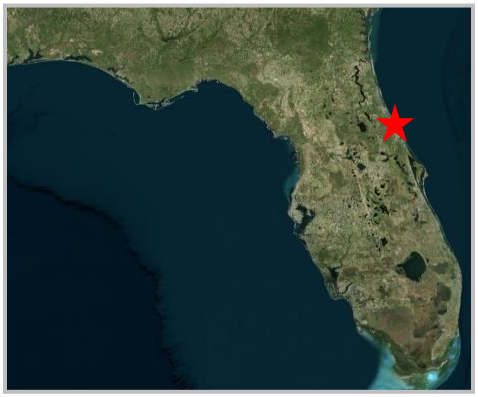
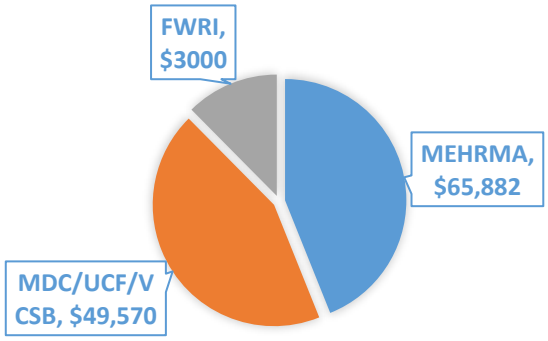
Marsh Madness, Mosquito Lagoon, Volusia, FL

FY 15/16



Project Partners and Funding



Project Description



Figure 1 High school students conduct monitoring of plants at the saltmarsh.

The Marsh Madness project is a result of a largescale saltmarsh restoration which was conducted on FWC property in New Smyrna Beach. The restoration (partially funded through MEHRMA) occurred in 2014 and the saltmarsh has been growing in terms of the spread of native plantings and the attraction of numerous species to the new habitat. FWC partnered with regional entities (Marine Discovery Center-MDC, University of Central Florida-UCF, and Volusia County School Board-VCSB) in order to survey the saltmarsh growth, species utilization, and various water quality parameters.

The project is run out of FWC's Mosquito Lagoon Marine Enhancement Center which houses MDC. During 2015/16, the partners created a region-wide biological sampling and surveying protocol of the new saltmarsh and reference areas by studying water quality, sediment elevation, various plant components, and use by animals (fish, invertebrates, birds) . To facilitate citizen science, citizens were trained to use the protocols and provided with the tools (eg, water quality testing kits) necessary. Biology teachers and students from the county AP programs were also trained in the field to use these data collection techniques. UCF conducted surveys at the new saltmarsh and reference sites to compare restoration techniques. FWRI conducted sediment sampling at various sites as part of a statewide wetland soil nutrient project. The partners also transplanted saltmarsh grasses from the new marsh to a recently restored one at a State Park in Ormond Beach. The

funding provided by MEHRMA for the Marsh Madness project was used as match towards the funds and time provided by the partners. As a result, the program is now established to continue in the future as the partners contribute data to the newly-created database.

Benefits

- Develop a regional hands-on citizen science monitoring network for estuarine wetlands
- Engage all Volusia County AP Biology teachers and students into citizen science
- Develop an ongoing database of survey data for the region
- Promote future research and monitoring projects by students of all ages
- Provide plants for regional restoration and teams for transplanting

FWC Strategic Plan Elements Supported

Theme 3, Goal 1: Further engage stakeholders and coordinate partnerships in the processes of developing and implementing conservation programs.

Strategies:

- Create and implement a common vision among partners and the FWC for improving and maintaining species populations and habitat, through interagency coordination, mutually supportive goals and initiatives
- Coordinate with partners on the development and implementation of hunting, fishing, boating and wildlife-viewing management actions

Acreage restored by habitat type (footprint)

This project entails citizen science monitoring of a 5-acre restored saltmarsh and reference sites:

- Hundreds of local citizens and AP Biology high school students engaged in science
- Transplantation of thousands of saltmarsh grass plugs to regional restoration areas

Acreage affected by restoration activities: 10 acres (includes newly restored and reference saltmarshes)



Figure 2: Monitoring of fishes in the restored marsh.



Figure 3: Partners thrilled with newly planted saltmarsh grasses at North Peninsula State Park.